Parent table	Child table	Foreign key	Cardinality	On delete	Meaning
USERS	USER_SKILLS	user_id	1 to many	CASCADE	A member chooses skills; at least one is needed to finish onboarding (checked in Flask).
SKILLS	USER_SKILLS	skill_id	1 to many	RESTRICT	Connects a member's skill to the shared skills list.
SKILLS	TASK_SKILLS	skill_id	1 to many	RESTRICT	Connects a task's skill need to the shared skills list.
TASKS	TASK_SKILLS	task_id	1 to many	CASCADE	Skills a task needs, each marked required or helpful.
TASK_CATEGORIES	TASKS	category_id	1 to many	RESTRICT	Each task has one type of help.
USERS	TASKS	creator_id	1 to many	RESTRICT	A help seeker creates many tasks.
TASKS	TASK_APPLICATIONS	task_id	1 to many	CASCADE	A task can receive offers from many volunteers.
USERS	TASK_APPLICATIONS	volunteer_id	1 to many	RESTRICT	A volunteer offers to help with many tasks. A withdrawn offer is deleted.
TASK_APPLICATIONS	REVIEWS	application_id	1 to 0..2	RESTRICT	One review per person per completed task (the volunteer and the help seeker), so at most two per application; admin can hide or restore.
USERS	REVIEWS	reviewer_id / reviewed_user_id	1 to many	RESTRICT	The member who writes a review and the member who receives it.
USERS	NOTIFICATIONS	user_id	1 to many	CASCADE	In app notifications (offer, accept, decline, close, withdraw, confirm, admin checks).
TASK_APPLICATIONS	NOTIFICATIONS	task_application_id	1 to many (optional)	SET NULL	Optional link to the offer the notification is about.
TASKS	CONVERSATIONS	task_id	1 to many	CASCADE	Messages about a task: API built and tested in Sprint 3; messaging page planned (Later).
CONVERSATIONS	CONVERSATION_MEMBERS	conversation_id	1 to many	CASCADE	People in a conversation.
USERS	CONVERSATION_MEMBERS	user_id	1 to many	CASCADE	Conversations a member belongs to.
CONVERSATIONS	MESSAGES	conversation_id	1 to many	CASCADE	Messages in a conversation.
USERS	MESSAGES	sender_id	1 to many	RESTRICT	Message sender; membership checked in Flask.
USERS	USER_VERIFICATIONS	user_id	1 to many	CASCADE	Optional verification with an ID document; the file is private and deleted after the decision.
USERS	REPORTS	reporter_id	1 to many	RESTRICT	A member can submit many reports and choose whether the team may contact them.
USERS	REPORTS	reported_user_id	1 to many (optional)	RESTRICT	Used only when target_type is user. For something else, other_details is used instead.
TASKS	REPORTS	task_id	1 to many (optional)	RESTRICT	Used only when target_type is task.
USERS	AUDIT_LOGS	actor_id	1 to many (optional)	SET NULL	Who did an action (profile change, withdraw, decisions); kept when a member is removed.
USERS	USER_VERIFICATIONS	reviewed_by	1 to many (optional)	SET NULL	Administrator who checked it; admin permission checked in Flask.
USERS	REPORTS	reviewed_by	1 to many (optional)	SET NULL	Administrator who checked it; admin permission checked in Flask.
USERS	REVIEWS	moderated_by	1 to many (optional)	SET NULL	Administrator who checked it; admin permission checked in Flask.

QuickVolunteer | 3NF entities

15 entities, matching database/schema.sql and the ERD (Sprint 3). Accounts are active from sign up. Verification is optional and uses an ID document that is deleted after the check. At least one skill is needed to finish onboarding. Deadlines are required; in person tasks need a postcode. Members type only a postcode: users.neighbourhood and tasks.location_text store the area name worked out from it (postcodes.io), so other members never see the postcode. When an administrator deletes a request, tasks.status becomes closed, so reports and reviews about it are kept (updated 25 Oct). Demo data: the database is filled with fictional demo data created with the help of AI tools; it contains no real personal data.

PK = primary key · FK = foreign key · PK / FK = composite key and foreign key · Later features not in the database: DBS checks, profile photos, preferred connections, volunteer levels, email or text notifications, organisations, group volunteering, community events, donations, a mobile app, changing language (users.preferred_language is already there for it). The accessibility tools bar needs no database table: choices are saved in the visitor's browser.

USERS

Attribute	Key	MySQL type	Null?	Constraint / reference
user_id	PK	BIGINT UNSIGNED	No	AUTO_INCREMENT
first_name	—	VARCHAR(50)	No	
last_name	—	VARCHAR(50)	No	Members see the initial only
email	—	VARCHAR(254)	No	UNIQUE
password_hash	—	VARCHAR(255)	No	Werkzeug hash only
phone	—	VARCHAR(30)	Yes	
postcode	—	VARCHAR(10)	Yes	Private; used for distance
neighbourhood	—	VARCHAR(80)	Yes	Area name from the postcode
preferred_language	—	VARCHAR(10)	Yes	
account_type	—	ENUM(help_seeker, volunteer, both, admin)	No	DEFAULT both
about_me	—	TEXT	Yes	Shown to other members
availability_notes	—	VARCHAR(255)	Yes	Days and times free
preferred_mode	—	ENUM(in_person, remote, any)	No	DEFAULT any
max_distance_miles	—	TINYINT UNSIGNED	Yes	CHECK 1 to 50
account_status	—	ENUM(pending, active, suspended)	No	DEFAULT active; 'pending' reserved, not used in V1
created_at	—	DATETIME	No	Automatic
updated_at	—	DATETIME	Yes	Last profile change

Active from sign up. Every profile change notifies the administrators.

USER_VERIFICATIONS

Attribute	Key	MySQL type	Null?	Constraint / reference
verification_id	PK	BIGINT UNSIGNED	No	AUTO_INCREMENT
user_id	FK	BIGINT UNSIGNED	No	FK → USERS.user_id; ON DELETE CASCADE
verification_method	—	ENUM(identity_document)	No	DEFAULT identity_document
evidence_reference	—	VARCHAR(500)	Yes	Private file name
status	—	ENUM(pending, approved, rejected)	No	DEFAULT pending
submitted_at	—	DATETIME	No	Automatic
reviewed_by	FK	BIGINT UNSIGNED	Yes	FK → USERS.user_id (administrator); ON DELETE SET NULL; Administrator
reviewed_at	—	DATETIME	Yes	
admin_notes	—	TEXT	Yes	

Optional, ID document only. The file is private and deleted after the decision.

SKILLS

Attribute	Key	MySQL type	Null?	Constraint / reference
skill_id	PK	BIGINT UNSIGNED	No	AUTO_INCREMENT
skill_name	—	VARCHAR(80)	No	UNIQUE
category	—	VARCHAR(80)	Yes	Skill group

12 skills in 5 groups, for example Shopping (Everyday help).

USER_SKILLS

Attribute	Key	MySQL type	Null?	Constraint / reference
user_id	PK / FK	BIGINT UNSIGNED	No	FK → USERS.user_id; ON DELETE CASCADE; composite primary key
skill_id	PK / FK	BIGINT UNSIGNED	No	FK → SKILLS.skill_id; ON DELETE RESTRICT; composite primary key
skill_level	—	ENUM(beginner, intermediate, advanced)	Yes	Optional
selected_at	—	DATETIME	No	Automatic

At least one skill is needed to finish onboarding (checked in Flask).

REVIEWS

Attribute	Key	MySQL type	Null?	Constraint / reference
review_id	PK	BIGINT UNSIGNED	No	AUTO_INCREMENT
application_id	FK	BIGINT UNSIGNED	No	FK → TASK_APPLICATIONS.application_id; ON DELETE RESTRICT
reviewer_id	FK	BIGINT UNSIGNED	No	FK → USERS.user_id; ON DELETE RESTRICT
reviewed_user_id	FK	BIGINT UNSIGNED	No	FK → USERS.user_id; ON DELETE RESTRICT
rating	—	TINYINT UNSIGNED	No	CHECK 1 to 5
comment	—	TEXT	Yes	
created_at	—	DATETIME	No	Automatic
moderation_status	—	ENUM(visible, hidden)	No	DEFAULT visible
moderated_by	FK	BIGINT UNSIGNED	Yes	FK → USERS.user_id (administrator); ON DELETE SET NULL; Administrator
moderated_at	—	DATETIME	Yes	
moderation_reason	—	TEXT	Yes	

UNIQUE (application_id, reviewer_id). Admin can hide or restore.

NOTIFICATIONS

Attribute	Key	MySQL type	Null?	Constraint / reference
notification_id	PK	BIGINT UNSIGNED	No	AUTO_INCREMENT
user_id	FK	BIGINT UNSIGNED	No	FK → USERS.user_id; ON DELETE CASCADE
task_application_id	FK	BIGINT UNSIGNED	Yes	FK → TASK_APPLICATIONS.application_id; ON DELETE SET NULL
title	—	VARCHAR(120)	No	
content	—	TEXT	No	
read_at	—	DATETIME	Yes	
created_at	—	DATETIME	No	Automatic

In app now; email or text message is a later feature.

CONVERSATIONS

Attribute	Key	MySQL type	Null?	Constraint / reference
conversation_id	PK	BIGINT UNSIGNED	No	AUTO_INCREMENT
task_id	FK	BIGINT UNSIGNED	No	FK → TASKS.task_id; ON DELETE CASCADE
created_at	—	DATETIME	No	Automatic

Messaging: API built and tested (Sprint 3); messaging page planned (Later).

CONVERSATION_MEMBERS

Attribute	Key	MySQL type	Null?	Constraint / reference
conversation_id	PK / FK	BIGINT UNSIGNED	No	FK → CONVERSATIONS.conversation_id; ON DELETE CASCADE; composite primary key
user_id	PK / FK	BIGINT UNSIGNED	No	FK → USERS.user_id; ON DELETE CASCADE; composite primary key
joined_at	—	DATETIME	No	Automatic
last_read_at	—	DATETIME	Yes	

MESSAGES

Attribute	Key	MySQL type	Null?	Constraint / reference
message_id	PK	BIGINT UNSIGNED	No	AUTO_INCREMENT
conversation_id	FK	BIGINT UNSIGNED	No	FK → CONVERSATIONS.conversation_id; ON DELETE CASCADE
sender_id	FK	BIGINT UNSIGNED	No	FK → USERS.user_id; ON DELETE RESTRICT
content	—	TEXT	No	
sent_at	—	DATETIME	No	Automatic

TASK_CATEGORIES

Attribute	Key	MySQL type	Null?	Constraint / reference
category_id	PK	BIGINT UNSIGNED	No	AUTO_INCREMENT
category_name	—	VARCHAR(80)	No	UNIQUE

8 types of help, for example Shopping and errands.

TASKS

Attribute	Key	MySQL type	Null?	Constraint / reference
task_id	PK	BIGINT UNSIGNED	No	AUTO_INCREMENT
creator_id	FK	BIGINT UNSIGNED	No	FK → USERS.user_id; ON DELETE RESTRICT
category_id	FK	BIGINT UNSIGNED	No	FK → TASK_CATEGORIES.category_id; ON DELETE RESTRICT
title	—	VARCHAR(160)	No	
description	—	TEXT	No	
location_text	—	VARCHAR(160)	Yes	Area name from the postcode (in person)
postcode	—	VARCHAR(10)	Yes	In person only
delivery_mode	—	ENUM(in_person, remote)	No	
estimated_minutes	—	SMALLINT UNSIGNED	No	CHECK 15 to 60
deadline_at	—	DATETIME	No	Required; entered by help seeker
status	—	ENUM(open, filled, completed, closed)	No	DEFAULT open
created_at	—	DATETIME	No	Automatic

CHECK: in person needs postcode and neighbourhood; remote has none.

TASK_SKILLS

Attribute	Key	MySQL type	Null?	Constraint / reference
task_id	PK / FK	BIGINT UNSIGNED	No	FK → TASKS.task_id; ON DELETE CASCADE; composite primary key
skill_id	PK / FK	BIGINT UNSIGNED	No	FK → SKILLS.skill_id; ON DELETE RESTRICT; composite primary key
importance	—	ENUM(required, helpful)	No	DEFAULT helpful

TASK_APPLICATIONS

Attribute	Key	MySQL type	Null?	Constraint / reference
application_id	PK	BIGINT UNSIGNED	No	AUTO_INCREMENT
task_id	FK	BIGINT UNSIGNED	No	FK → TASKS.task_id; ON DELETE CASCADE
volunteer_id	FK	BIGINT UNSIGNED	No	FK → USERS.user_id; ON DELETE RESTRICT; UNIQUE (task_id, volunteer_id)
status	—	ENUM(pending, accepted, rejected, completed)	No	DEFAULT pending
applied_at	—	DATETIME	No	Automatic
accepted_at	—	DATETIME	Yes	
volunteer_completed_at	—	DATETIME	Yes	
help_seeker_completed_at	—	DATETIME	Yes	

Completed when both confirm. A withdrawn offer is deleted.

REPORTS

Attribute	Key	MySQL type	Null?	Constraint / reference
report_id	PK	BIGINT UNSIGNED	No	AUTO_INCREMENT
reporter_id	FK	BIGINT UNSIGNED	No	FK → USERS.user_id; ON DELETE RESTRICT
target_type	—	ENUM(user, task, other)	No	
reported_user_id	FK	BIGINT UNSIGNED	Yes	FK → USERS.user_id; ON DELETE RESTRICT; When target is user
task_id	FK	BIGINT UNSIGNED	Yes	FK → TASKS.task_id; ON DELETE RESTRICT; When target is task
other_details	—	TEXT	Yes	When target is other
reason	—	TEXT	No	
contact_allowed	—	BOOLEAN	No	DEFAULT TRUE
reviewed_by	FK	BIGINT UNSIGNED	Yes	FK → USERS.user_id (administrator); ON DELETE SET NULL; Administrator
status	—	ENUM(open, reviewing, closed)	No	DEFAULT open
created_at	—	DATETIME	No	Automatic
reviewed_at	—	DATETIME	Yes	

CHECK: exactly one target; other needs other_details.

AUDIT_LOGS

Attribute	Key	MySQL type	Null?	Constraint / reference
audit_log_id	PK	BIGINT UNSIGNED	No	AUTO_INCREMENT
actor_id	FK	BIGINT UNSIGNED	Yes	FK → USERS.user_id; ON DELETE SET NULL
action	—	VARCHAR(100)	No	
entity_type	—	VARCHAR(50)	No	
entity_id	—	BIGINT UNSIGNED	Yes	
created_at	—	DATETIME	No	Automatic

QuickVolunteer | Normalisation as tables

Example: one help request and its applications from UNF to 3NF. Sign-up activates accounts; matching needs at least one selected skill. Demo data: the database is filled with fictional demo data created with the help of AI tools; it contains no real personal data.

UNF

TASK_REQUEST_UNF

Attribute	Key / MySQL type
task_id	PK
task_title	
creator_id / creator_name	
category_id / category_name	
applications	Repeating list: volunteer + status + date

→

1NF

TASK_APPLICATION_1NF

Attribute	Key / MySQL type
task_id	PK*
task_title	
creator_id / creator_name	
category_id / category_name	
volunteer_id / volunteer_name	PK*
application_status	
applied_at	

One row per task–volunteer pair. Candidate key: (task_id, volunteer_id). All values are atomic; task and user details still repeat.

→

2NF

TASKS (2NF)

Attribute	Key / MySQL type
task_id	PK
title	
creator_id	FK
category_id	FK
category_name	

USERS (2NF)

Attribute	Key / MySQL type
user_id	PK
first_name	
last_name	
email	

TASK_APPLICATIONS (2NF)

Attribute	Key / MySQL type
task_id	PK / FK
volunteer_id	PK / FK
status	
applied_at	

Partial dependencies removed: task attributes depend on task_id; user attributes on user_id; application attributes on the full pair. category_id → category_name remains.

3NF: final schema, 15 entities (matches database/schema.sql)

Members choose at least one skill. Suggestions use skills first, then the distance between postcode districts. If role selection is skipped, account_type is both. Verification is optional with an ID document. Full rules are in the 3NF Entities sheet.

USERS

Attribute	Key / MySQL type
user_id	PK · BIGINT UNSIGNED
first_name	VARCHAR(50)
last_name	VARCHAR(50)
email	VARCHAR(254)
password_hash	VARCHAR(255)
phone	VARCHAR(30) · NULL
postcode	VARCHAR(10) · NULL
neighbourhood	VARCHAR(80) · NULL
preferred_language	VARCHAR(10) · NULL
account_type	ENUM(help_seeker, volunteer, both, admin)
about_me	TEXT · NULL
availability_notes	VARCHAR(255) · NULL
preferred_mode	ENUM(in_person, remote, any)
max_distance_miles	TINYINT UNSIGNED · NULL
account_status	ENUM(pending, active, suspended)
created_at	DATETIME
updated_at	DATETIME · NULL

USER_VERIFICATIONS

Attribute	Key / MySQL type
verification_id	PK · BIGINT UNSIGNED
user_id	FK · BIGINT UNSIGNED
verification_method	ENUM(identity_document)
evidence_reference	VARCHAR(500) · NULL
status	ENUM(pending, approved, rejected)
submitted_at	DATETIME
reviewed_by	FK · BIGINT UNSIGNED · NULL
reviewed_at	DATETIME · NULL
admin_notes	TEXT · NULL

SKILLS

Attribute	Key / MySQL type
skill_id	PK · BIGINT UNSIGNED
skill_name	VARCHAR(80)
category	VARCHAR(80) · NULL

USER_SKILLS	
Attribute	**Key / MySQL type**
user_id	PK / FK · BIGINT UNSIGNED
skill_id	PK / FK · BIGINT UNSIGNED
skill_level	ENUM(beginner, intermediate, advanced) · NULL
selected_at	DATETIME

TASK_CATEGORIES	
Attribute	**Key / MySQL type**
category_id	PK · BIGINT UNSIGNED
category_name	VARCHAR(80)

TASKS	
Attribute	**Key / MySQL type**
task_id	PK · BIGINT UNSIGNED
creator_id	FK · BIGINT UNSIGNED
category_id	FK · BIGINT UNSIGNED
title	VARCHAR(160)
description	TEXT
location_text	VARCHAR(160) · NULL
postcode	VARCHAR(10) · NULL
delivery_mode	ENUM(in_person, remote)
estimated_minutes	SMALLINT UNSIGNED
deadline_at	DATETIME
status	ENUM(open, filled, completed, closed)
created_at	DATETIME

TASK_SKILLS	
Attribute	**Key / MySQL type**
task_id	PK / FK · BIGINT UNSIGNED
skill_id	PK / FK · BIGINT UNSIGNED
importance	ENUM(required, helpful)

TASK_APPLICATIONS	
Attribute	**Key / MySQL type**
application_id	PK · BIGINT UNSIGNED
task_id	FK · BIGINT UNSIGNED
volunteer_id	FK · BIGINT UNSIGNED
status	ENUM(pending, accepted, rejected, completed)
applied_at	DATETIME
accepted_at	DATETIME · NULL
volunteer_completed_at	DATETIME · NULL
help_seeker_completed_at	DATETIME · NULL

REVIEWS	
Attribute	**Key / MySQL type**
review_id	PK · BIGINT UNSIGNED
application_id	FK · BIGINT UNSIGNED
reviewer_id	FK · BIGINT UNSIGNED
reviewed_user_id	FK · BIGINT UNSIGNED
rating	TINYINT UNSIGNED
comment	TEXT · NULL
created_at	DATETIME
moderation_status	ENUM(visible, hidden)
moderated_by	FK · BIGINT UNSIGNED · NULL
moderated_at	DATETIME · NULL
moderation_reason	TEXT · NULL

NOTIFICATIONS	
Attribute	**Key / MySQL type**
notification_id	PK · BIGINT UNSIGNED
user_id	FK · BIGINT UNSIGNED
task_application_id	FK · BIGINT UNSIGNED · NULL
title	VARCHAR(120)
content	TEXT
read_at	DATETIME · NULL
created_at	DATETIME

CONVERSATIONS	
Attribute	**Key / MySQL type**
conversation_id	PK · BIGINT UNSIGNED
task_id	FK · BIGINT UNSIGNED
created_at	DATETIME

CONVERSATION_MEMBERS	
Attribute	**Key / MySQL type**
conversation_id	PK / FK · BIGINT UNSIGNED
user_id	PK / FK · BIGINT UNSIGNED
joined_at	DATETIME
last_read_at	DATETIME · NULL

MESSAGES	
Attribute	**Key / MySQL type**
message_id	PK · BIGINT UNSIGNED
conversation_id	FK · BIGINT UNSIGNED
sender_id	FK · BIGINT UNSIGNED
content	TEXT
sent_at	DATETIME

REPORTS	
Attribute	**Key / MySQL type**
report_id	PK · BIGINT UNSIGNED
reporter_id	FK · BIGINT UNSIGNED
target_type	ENUM(user, task, other)
reported_user_id	FK · BIGINT UNSIGNED · NULL
task_id	FK · BIGINT UNSIGNED · NULL
other_details	TEXT · NULL
reason	TEXT
contact_allowed	BOOLEAN
reviewed_by	FK · BIGINT UNSIGNED · NULL
status	ENUM(open, reviewing, closed)
created_at	DATETIME
reviewed_at	DATETIME · NULL

AUDIT_LOGS	
Attribute	**Key / MySQL type**
audit_log_id	PK · BIGINT UNSIGNED
actor_id	FK · BIGINT UNSIGNED · NULL
action	VARCHAR(100)
entity_type	VARCHAR(50)
entity_id	BIGINT UNSIGNED · NULL
created_at	DATETIME